



ElectronicsTutorials

ALTERNATING CURRENT TRANSFORMERS

For Students, Professionals
and Beyond

eBook 10

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TABLE OF CONTENTS

1. Introduction	1
2. A Transformers Turns Ratio (TR)	2
3. Ideal Transformer Action.	3
4. Transformer on No-Load	4
5. The Transformer On-Load	5
6. Transformer Voltage Regulation	6
7. Transformer Rating	7
8. Transformer Dot Orientation	7
9. Multiple Winding Voltage Transformers	8
10. The Autotransformer	9
11. 3-Phase Transformers.	10

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1. INTRODUCTION

Transformers are static electromagnetic devices that require no moving parts to operate, but instead are able to transfer electrical energy from one circuit to another in a quick and efficient way without any direct electrical connection.

Transformers work in alternating current (AC) circuits and are capable of either increasing or decreasing the voltage and current levels of their supply, without modifying its frequency, or the amount of electrical power being transferred from one coil winding to another. That is, whatever frequency is on the primary will appear on the secondary.

As such, transformers get their name from the fact that they can “transform” one voltage or current level into another and are used in many different types of applications such as electrical power generation, transmission and electronic power supply systems.

The electrical operation of a transformer can be best explained by the means of electromagnetic theory consisting of two individual coils placed very close together so that the flux generated by the driving coil also links with the other coil. Thus a transformer requires at least two electrical coils, or windings to operate plus a magnetic core.

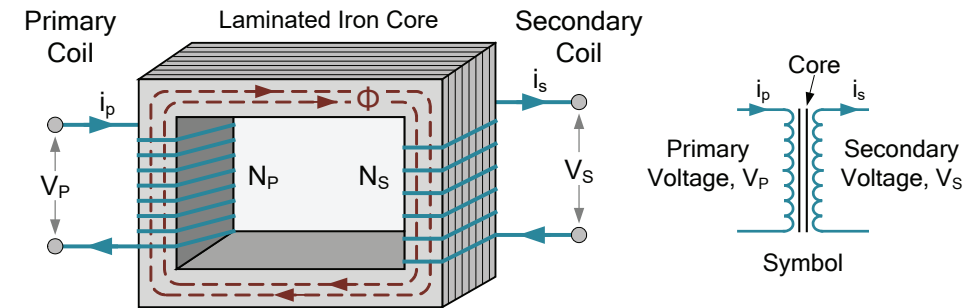
A single phase voltage transformer basically consists of two electrical coils of wire wound around a common core. The main driving coil (or winding) connected to the supply voltage is called the **PRIMARY COIL**. While the second receiving coil which is connected to the load is called the **SECONDARY COIL**. The primary and secondary coils are closely wound together onto a single laminated iron core, as shown in Figure 1.

Transformers are Electromagnetic devices that transfer electrical energy from one circuit to another

The “primary” side of a transformer is commonly defined as being the input side that receives electrical power from an external source. The “secondary” side of a transformer is defined as being the output side which delivers electrical power to a connected load.

For a single-phase transformer, the primary is usually the side with the higher voltage value attached, but multiple secondary windings can be used to obtain different voltages.

FIGURE 1. TRANSFORMER CONSTRUCTION AND SYMBOL



Then we can see from Figure 1. above, that the two wound coils are not in direct electrical contact with each other but instead are wrapped together around a common closed magnetic iron circuit called the “**core**”.

This soft iron core is not solid but made up of individual laminations connected together to help reduce the core’s losses and therefore transfer the electrical energy by electromagnetic induction (transformer action) from the primary side to the secondary side more efficiently.

Then the two coil windings are electrically isolated from each other but are magnetically linked through the common core allowing electrical power to be transferred from one coil to the other. That is a transformer provides electrical isolation between the two coil windings while still supplying power.

When a transformer is used to “increase” the voltage on its secondary winding with respect to the primary, it is called a **Step-up transformer**. When a transformer is used to “decrease” the voltage on the secondary winding with respect to the primary it is called a **Step-down transformer**.

However, if the transformer supplies the same voltage on its secondary as is applied to its primary winding, that is its output is identical with respect to voltage, current and power transferred. Then this type of transformer is commonly called a 1-to-1 isolation transformer.

2. A TRANSFORMERS TURNS RATIO (TR)

As we saw in eBook 9 about electromagnetism, when an electric current passed through the primary winding, a magnetic field is developed around it which induces a voltage into the secondary coil winding via the core. The difference in voltage (or current) between the primary and the secondary windings is achieved by changing the number of coil turns within the primary winding, N_p compared to the number of coil turns of wire wound onto the secondary winding, N_s , or vice versa.

As the transformer is basically a linear device, a ratio now exists between the number of turns of the primary coil divided by the number of turns of the secondary coil. This ratio, called the ratio of transformation, but more commonly known as a transformers “turns ratio”, or transformer ratio, (TR). This turns ratio value dictates the operation of the transformer and the corresponding voltage and current values available on the secondary winding and thus to the connected load.

The turns ratio of a transformer compares the number of turns on each winding and is presented as two numbers separated by a colon, such as 3:1 (3-to-1). Note that it has no units as it's a ratio. This means that when there is 3 volts on the primary winding there will be 1 volt on the secondary winding, 3 volts-to-1 volt. So we can see that if the ratio between the number of turns changes, the resulting voltages must also change by the same ratio amount.

Transformers are all about “ratios”. The ratio of the primary to the secondary, the ratio of the input to the output. The turns ratio of any given transformer will also be the same as its voltage ratio. In other words, for a transformer “turns ratio = voltage ratio”. The actual number of turns of wire on any winding is generally not important, just the turns ratio and this relationship is given as:

$$\text{Turns Ratio, (TR)} = \frac{N_p}{N_s} = \frac{V_p}{V_s} = \frac{I_s}{I_p}$$

Where: N_p is the *Number of Primary* windings, N_s is the *Number of Secondary* windings, V_p is the *Primary Voltage*, V_s is the *Secondary Voltage*, I_p is the *Primary Current* and I_s is

the *Secondary Current*. Note also that $N_p/N_s = V_p/V_s$ which represents the voltage ratio. Whereas: $N_p/N_s = I_s/I_p$ represents the current ratio, which is inversely proportional to both the voltage and the number of turns.

Then in order to maintain a balanced power level across the transformers windings from the primary input to the secondary output, if the voltage is stepped up, the current must be stepped down and vice versa. In other words, “higher voltage — lower current” or “lower voltage — higher current”.

Transformers are all about Ratios between the input and output

As a transformer is about ratio's, the relationships between the number of turns on the primary and secondary, the voltage across each winding, and the current through the windings, can be found by rearranging the transformer ratio equation to find the value of any unknown voltage, (V) current, (I) or number of turns, (N).

$$\begin{aligned} V_p &= \frac{V_s N_p}{N_s} = \frac{V_s I_s}{I_p}, & V_s &= \frac{V_p N_s}{N_p} = \frac{V_p I_p}{I_s} \\ N_p &= \frac{V_p N_s}{N_p} = \frac{N_s I_s}{I_p}, & N_s &= \frac{V_s N_p}{V_p} = \frac{N_p I_p}{I_s} \\ I_p &= \frac{V_s I_s}{V_p} = \frac{N_s I_s}{N_p}, & I_s &= \frac{V_p I_p}{V_s} = \frac{N_p I_p}{N_s} \end{aligned}$$

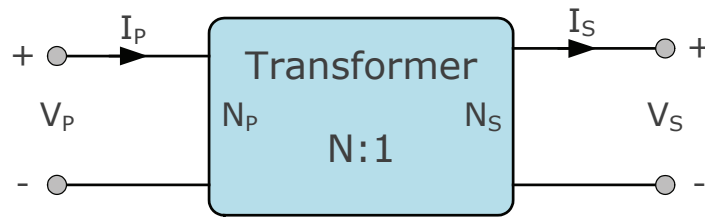
If the secondary voltage is to be lower or less than the input voltage, (step-down transformer) then there must be less turns on the secondary side giving a turns ratio of N:1, (N-to-1) where N represents the turns ratio number.

Likewise, if it is required that the secondary voltage is to be greater or higher than the primary, (step-up transformer) then the number of secondary windings must be more giving a turns ratio of 1:N, (1-to-N). Then the primary to secondary ratio remains the same regardless of the variations in primary input voltage.

Therefore, if the turns ratio is less than unity, $TR < 1$ then N_s is greater than N_p and the transformer is classed as a step-up transformer. If the turns ratio is greater than unity, that is: $TR > 1$, then N_p is greater than N_s , so the transformer will be classed as a step-down transformer. If the turns ratio is equal to unity, that is $TR = 1$, then both the primary and secondary have the same number of coil turns so therefore the voltages and currents will be the same for both the primary and secondary windings.

Then we can represent the ideal transformer in block diagram form as shown in Figure 2.

FIGURE 2. BASIC REPRESENTATION OF A TRANSFORMER



Note that the order of the numbers when expressing a transformer's turns ratio value is very important as the turns ratio 3:1 expresses a very different transformer relationship and output voltage than one in which the turns ratio is given as: 1:3. Thus a 240V transformer with a turns ratio of 3:1 will produce a secondary voltage of 80 volts.

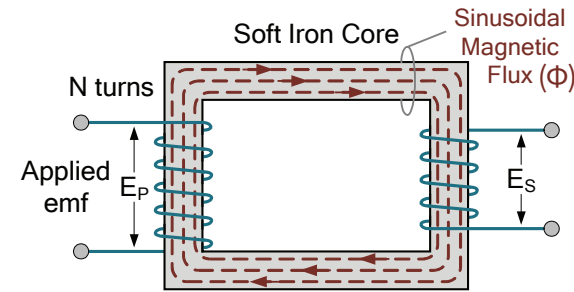
Also, a step-down transformer can also be used as a step-up transformer by simply reversing its connections and making the low voltage winding its primary, and vice versa as long as the transformer is operated within its original VA design rating with regards to voltage and current values.

3. IDEAL TRANSFORMER ACTION

We have said previously that a transformer basically consists of two coils wound around a common soft iron core and that the number of coil turns on the secondary winding compared to the primary winding, the turns ratio, affects the amount of voltage available from the secondary coil. But if the two windings are electrically isolated from each other, how is this secondary voltage produced.

When a sinusoidal AC voltage is applied to the primary coil of a transformer, current flows through the coil which in turn sets up a magnetic field around itself acting like an electromagnet according to Faraday's Law of electromagnetic induction. The strength of the magnetic field increases and decreases sinusoidally as the primary current flow alternates between zero and its maximum value, given as $d\Phi/dt$.

FIGURE 3. TRANSFORMER ACTION



As the magnetic flux created by the primary winding expands outward from the coil, the soft iron core forms a closed path for the flux to circulate around concentrating the magnetic flux.

This magnetic flux links the turns of both windings as it increases and decreases in opposite directions under the influence

of the sinusoidal AC supply.

However, the strength of the magnetic field (magnetic flux density, B) induced into the soft iron core depends upon the amount of current and the number of primary turns. Thus as the primary current is reduced, the magnetic field strength circulating around the core also reduces.

As the magnetic lines of flux flow around the core, they pass through the turns of the secondary winding, causing an emf to be induced into the secondary coil. The amount of secondary induced voltage will be determined by: $N \cdot d\Phi/dt$ (Faraday's Law), where N is the number of coil turns. This secondary induced voltage will have the same angular frequency as the primary winding voltage.

Then we can see that the same voltage is induced in each coil turn of both windings because the same magnetic flux links the turns of both the windings together. As a result, the total induced voltage in each winding is directly proportional to the number of turns in that winding. However, the peak amplitude of the output voltage available on the secondary winding will be reduced if the magnetic losses of the core are high.

If we want the primary coil to produce a stronger magnetic field to overcome the cores magnetic losses, we can either send a larger current through the primary coil, or keep the same current flowing, and instead increase the number of coil turns of the winding. The product of amperes times turns ($I \times N$) is called the “ampere-turns”, which determines the magnetising force, (H) of the primary coil.

So assuming we have a simple transformer with one single turn on the primary, and only one single turn on the secondary. If one volt is applied to the one turn of the primary coil, assuming no losses, enough current must flow and enough magnetic flux must be generated to induce one volt in the single turn of the secondary. That is, each winding supports the same number of volts per turn.

As the magnetic flux changes sinusoidally, the instantaneous value of the magnetic flux in the core is given as: $\Phi = \Phi_{MAX} \sin(\omega t)$ webers. Then the maximum value of the flux in the core varies between $+\Phi_{MAX}$ to $-\Phi_{MAX}$ in each half cycle. That is $T/2$ or π radians.

Therefore, the instantaneous value of the induced emf per coil turn is given as $d\Phi/dt$ volts. Since $\omega = 2\pi f$, the basic relationship between instantaneously induced emf, (E_t) in a coil of N turns is given by the equation relating the coil turns times rate of change of flux as a result of the supply frequency, f in Hertz:

$$E_{P(t)} = N_P \frac{d\Phi}{dt} \text{ volts}$$

Therefore:

$$E_{P(t)} = N_P \frac{d\Phi}{dt} = N_P \times 2\pi f \Phi_{MAX} \times \sin\left(2\pi f t + \frac{\pi}{2}\right)$$

As the maximum value of induced emf per turn equals: $2\pi f \Phi_{MAX}$

$$E_{P(MAX)} = N_P \times 2\pi f \Phi_{MAX}$$

$$E_{P(RMS)} = N_P \frac{2\pi f}{\sqrt{2}} \Phi_{MAX} = \frac{2\pi}{\sqrt{2}} N_P f \Phi_{MAX} = 4.44 N_P f \Phi_{MAX}$$

Clearly then the RMS (Root Mean Squared) value of the induced emf in the secondary winding will be:

$$E_{S(RMS)} = N_S \frac{2\pi f}{\sqrt{2}} \Phi_{MAX} = 4.44 N_S f \Phi_{MAX}$$

These equations are known commonly as the Transformer EMF Equation. For the primary winding emf, N will be the number of primary turns, (N_P) and for the secondary winding, N will be the number of secondary turns, (N_S).

As we can see from the above transformer equations, transformers require an alternating magnetic flux to operate correctly, transformers cannot therefore be used to transform or supply DC voltages or currents, since the magnetic field must be changing to induce a voltage in the secondary winding. In other words, transformers DO NOT operate on steady state DC voltages, only alternating, sinusoidal or pulsating voltages.

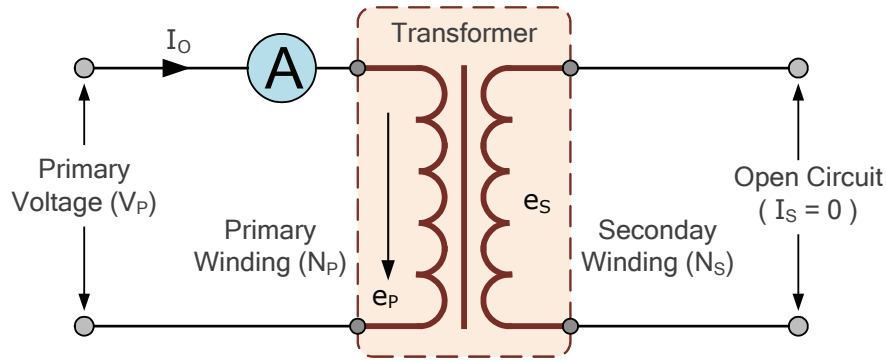
4. TRANSFORMER ON NO-LOAD

We have seen previously that a transformer is a two-coil device that uses electromagnetic induction to pass an AC signal from its primary (input) winding to its secondary (output) winding, while at the same time providing electrical isolation between the two.

But the primary winding is also a coil of wire which creates a magnetic flux around itself when a current flows through it, acting a bit like an inductor. Thus being a wound coil, the primary winding has both resistance and reactance. Therefore an electric current will always flow in the primary winding whether there is a secondary load connected or not.

A transformer is said to be in a condition of “**no-load**” when its secondary side winding is open-circuited. That is nothing is connected to the secondary terminals, so the transformer loading is zero. Consider the transformer of Figure 4.

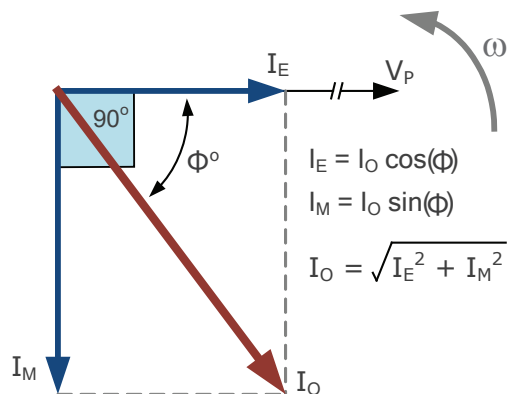
FIGURE 4. TRANSFORMER NO-LOAD CONDITION



When an alternating voltage supply is connected to the primary winding of a transformer, the ammeter (A) will indicate a small open-circuit current (I_o) flowing through the primary coil winding even though the secondary circuit is open-circuited. This no-load primary current is due to the presence of the primary supply voltage which in turn creates an alternating magnetic flux within the core.

The current which flows through the energised winding is called an “exciting current” or “magnetising current,” consisting of a real component and a reactive component as shown in Figure 5.

FIGURE 5. PRIMARY NO-LOAD CURRENT PHASOR DIAGRAM



The real or exciting current, (I_E) is “in-phase” with the supply voltage and supplies the no-load losses (eddy current and hysteresis) in the core independent of load.

The reactive or magnetising current, (I_M) lags the primary voltage by 90° so is “out-of-phase” with the applied voltage. This reactive component delivers no power but sets up the magnetic flux that stores energy within the windings inductance.

Note that this no-load primary current, I_o is very small compared to the transformers normal full-load current at the rated VA value.

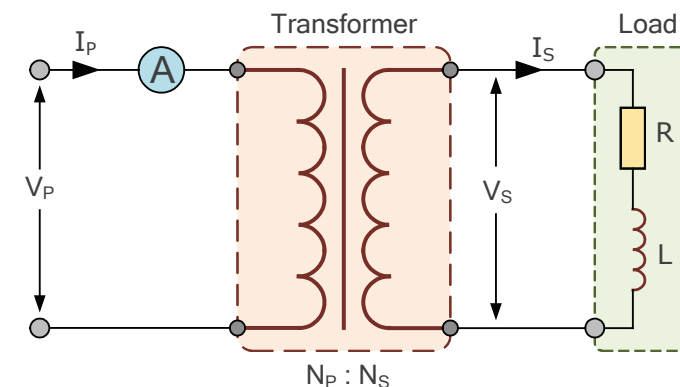
Also due to the iron losses present within the core, as well as the copper losses in the primary winding, I_o does not lag behind the supply voltage, V_p by exactly 90° ($\cos\Phi = 0$). Instead there will still be a large phase angle resulting in a poor power factor. If an open-circuited secondary transformer has an exciting current of 2 amperes and a magnetising current of 5 amperes. The no-load primary current would be 5.4 amperes at 0.93 p.f.

5. THE TRANSFORMER ON-LOAD

Having seen that a no-load primary current flows when the secondary winding is open-circuited, and as such is independent of load. Now let us look at what happens to the transformer when it is fully loaded to its rated value. That is some electrical circuit is connected to the secondary terminals of the transformer.

When an electrical load is connected to the secondary winding of a transformer as shown in Figure 6, the transformer loading is no longer zero, so a secondary current will flow to the connected load at a value determined by the connected impedance. The secondary current is a direct result of the induced voltage, set up by the magnetic flux created in the core by the primary current.

FIGURE 6. TRANSFORMER WITH A SECONDARY LOAD ATTACHED



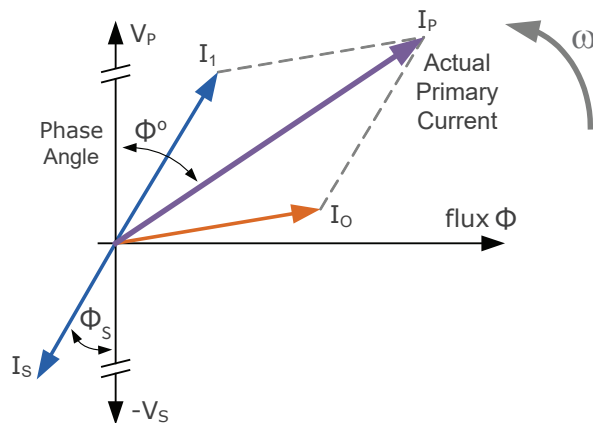
The magnitude of the secondary current, I_s is determined by the impedance characteristics of the connected load. This secondary current creates a self-induced secondary magnetic field, Φ_s in the transformer core which flows in the exact opposite direction to the main primary magnetic field, Φ_p .

These two magnetic fields oppose each other resulting in a combined magnetic field of less magnetic strength than the single field produced by the primary winding alone when the secondary circuit was previously open circuited.

Since $E_p = 4.44N_p f \Phi_p$ volts, any change in the magnetic flux within the core will change the value of E_p . So the combined magnetic field reduces the back EMF of the primary winding causing the primary current, I_p to increase slightly. The primary current continues to increase until the core's magnetic field is back at its original strength, and E_p is very nearly equal to V_p .

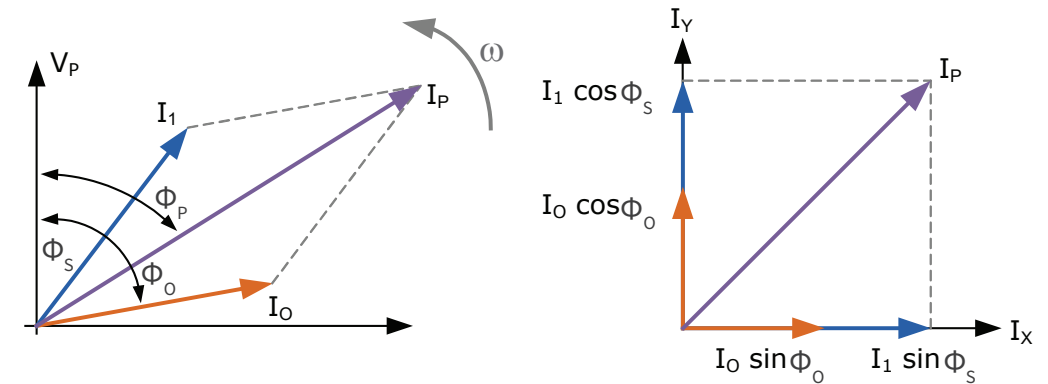
For a transformer to operate correctly, a balanced condition must always exist between the primary and secondary magnetic fields, resulting in a balanced power transfer on both the primary and secondary sides. Then the total current, I_p drawn from the supply by the primary winding will be the vector sum of the no-load current, I_o and the additional supply current, I_1 as a result of the secondary transformer loading as shown in Figure 7.

FIGURE 7. TRANSFORMER LOADING CURRENT



If we are given the secondary current, I_s and the no-load primary current, I_o and the turns ratio of N_p/N_s . We can calculate the exact primary current, I_p as from the previous equations: $I_p N_p = I_s N_s$. Therefore: $I_1 = (I_s N_s)/N_p$.

FIGURE 8. PRIMARY CURRENT DUE TO LOADING



Horizontal Component, $I_X = I_O \sin \Phi_o + I_1 \sin \Phi_s$

Vertical Component, $I_Y = I_O \cos \Phi_o + I_1 \cos \Phi_s$

$$I_P = \sqrt{I_X^2 + I_Y^2} \text{ and Power Factor, } pF = \cos \phi = \frac{I_Y}{I_P}$$

6. TRANSFORMER VOLTAGE REGULATION

The voltage regulation of a transformer is defined as the change in secondary terminal voltage when the transformer's secondary load is reduced from full-load to no-load while the input is at its maximum, i.e. full-load applied while the primary supply voltage is held constant. Regulation determines the voltage drop (or increase) that occurs inside the transformer as the load voltage becomes too low as a result of the transformer's loading being too high which therefore affects its performance and efficiency.

Voltage regulation is expressed as a percentage (or per unit) of the no-load voltage. Then if E_s represents the no-load secondary voltage and V_s represents the full-load secondary voltage, the percentage regulation of a transformer is defined as being:

$$\text{Percentage Regulation} = \frac{E_s(\text{No-load}) - V_s(\text{Full-load})}{E_s(\text{No-load})} \times 100\%$$

So for example, if a transformer has a secondary voltage of 100 volts at no-load, and the voltage drops to 95 volts at full load current, the regulation would be 5%. Note that the value of $E_s - V_s$ will depend upon the internal impedance of the secondary winding.

Also, a transformers voltage regulation value can be either positive or negative in value. That is with the no-load voltage as reference, the change down in regulation as the load is applied, or with the full-load as reference and the change up in regulation as the load is reduced or removed.

7. TRANSFORMER RATING

Transformers have an electrical rating based upon the continuous power output they are capable of supplying at the rated voltage, current and frequency without deterioration or overheating. Commonly, a transformers rating is given in volt-amperes, (VA) or multiples of volt-amperes. For example, 1000 VA is equal to 1kVA, or 1000000 VA (1000kVA) is equal to 1MVA.

Transformers are commonly rated in Volt-amperes (VA) rather than in electrical power (watts) since no power is actually dissipated by the transformer. The transformers volt-amperes rating defines the amount of power that it can safely deliver to a resistive load having a power factor of one (1). Volt-amperes is the product of the RMS voltage times the RMS current and not power in watts as power $(W) = VA \times \text{power factor}$, $(W = VA \times \cos\theta)$.

Also, the internal I^2R copper losses (winding resistance loss) of both its primary and secondary windings depends on current, while losses in the core due to magnetic and eddy-current losses are a result of voltage.

The amount of copper loss is proportional to the square of the load current, while the core's hysteresis and eddy-current losses vary with the input voltage and frequency. Hence any internal losses are a result of volt-ampere values and not phase angle.

As a result, neither of these losses depends on the power being consumed by the connected load and not any phase angle between them. It is the impedance of the connected load which determines the load power factor, and therefore the watts consumed. Thus transformers are rated in volt-amperes (VA) or kilovolt-amperes (kVA) and not in watts (W), or kilo-watts (kW).

8. TRANSFORMER DOT ORIENTATION

Thus far we have considered transformers with one single primary winding and one single secondary winding. But multi-winding transformers which contain more than one primary or more than one secondary winding on a common laminated core also commonly exist as shown in Figure 9.

FIGURE 9. A MULTI-WINDING TRANSFORMER

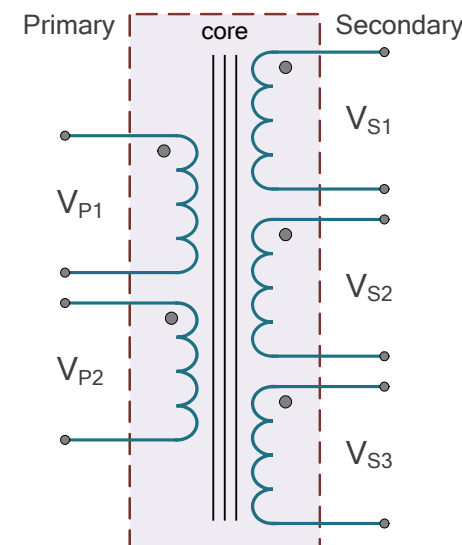


Figure 9. shows a simple example of a typical multiple winding transformer which has a number of different primary and secondary windings on the same laminated core.

The primary windings can be used individually or connected together in series to operate the transformer from a much higher voltage supply. The secondary windings can also be connected together in various configurations producing a higher voltage or current supply if needed.

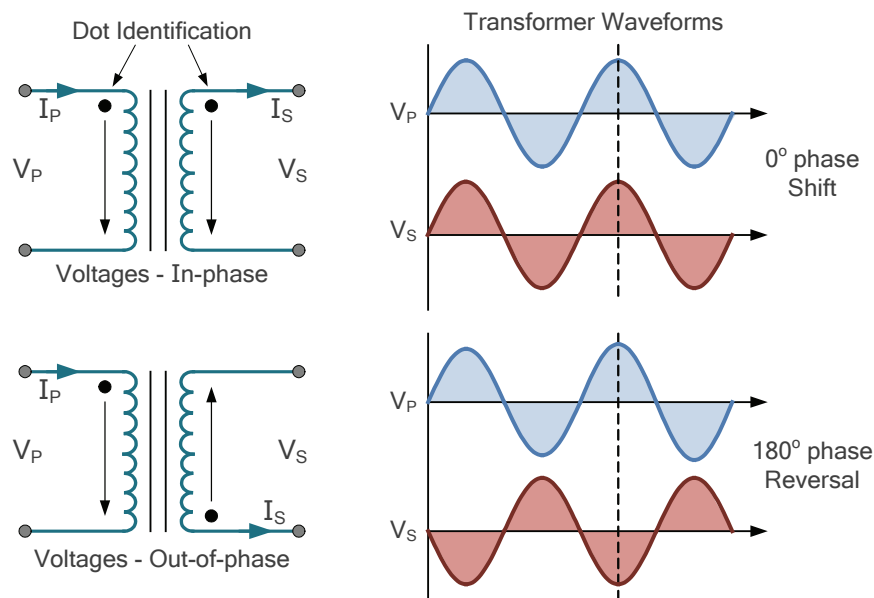
But we cannot just simply take a laminated core and wrap different coil windings around it. We could but we may find that the secondary voltages and currents may be out-of-phase with

that of the primary voltage and current, or that the windings short each other out.

One simple and effective method of identifying the orientation or wound direction of a transformers primary or secondary side coils is to use what is called the “dot convention”. A dot convention is used with transformer coils to indicate the relative polarity of a transformers windings so that multiple coil windings on the same core will have a distinct orientation with respect to the other without having to show their actual construction.

Then the windings of a transformer are wound and connected so that the correct phase relations exist between the winding voltages with the transformer’s polarity being defined as the relative polarity of the secondary voltage with respect to the primary voltage. Whether that is “in-phase” or 180° “out-of-phase” as shown in Figure 10.

FIGURE 10. TRANSFORMER CONSTRUCTION USING DOT ORIENTATION



The first transformer of Figure 10. shows its two “dots” side by side on the two windings. The current leaving the secondary dot is “in-phase” with the current entering the primary

side dot. Thus the polarities of the voltages at the dotted ends are also in-phase so when the voltage is positive at the dotted end of the primary coil, the voltage across the secondary coil is also positive at the dotted end.

The second transformer of Figure 10. shows the two dots at opposite ends of the windings which means that the primary and secondary coil windings of the transformer are wound in opposite directions. The result of this is that the current leaving the secondary dot is 180° out-of-phase with the current entering the primary dot. Therefore, the polarities of the voltages at the dotted ends are also out-of-phase.

9. MULTIPLE WINDING VOLTAGE TRANSFORMERS

The principal of operation of a multiple winding voltage transformer is no different from that of an ordinary transformer. The primary and secondary voltages, currents and turns ratios are all calculated exactly the same. The difference this time is that we need to pay special attention to the voltage polarities of each coil winding, as well as the dot convention marking the polarity of the winding when they are connected together.

FIGURE 11. SERIES CONNECTED SECONDARY WINDINGS

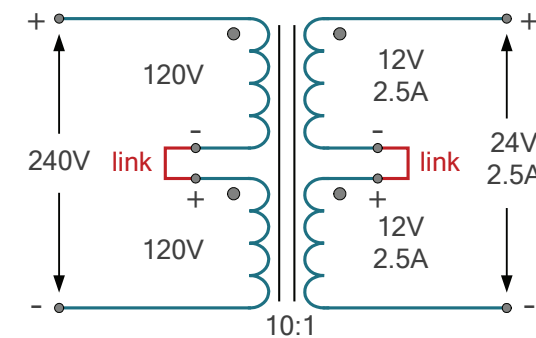


Figure 11. shows the two identically rated 120V primary windings connected together in series across a 240V supply.

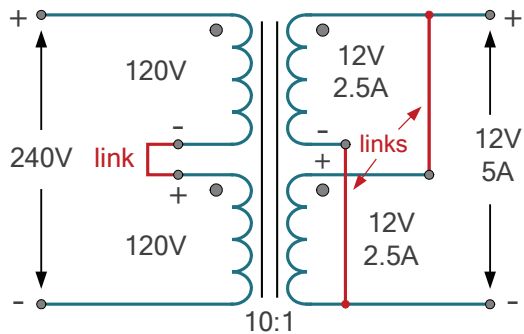
Thus half the supply voltage, namely 120V, is dropped across each winding and the same primary current flows through both.

The two secondary windings rated at 12V, 2.5A each are connected in series with the secondary terminal voltage being the sum

of the two individual winding voltages giving 24 Volts.

As the two windings are connected in series, the same amount of current flows through each winding, then the secondary current is the same at 2.5 Amps. So for a series connected secondary, the output of Figure 11. is rated at 24 Volts, 2.5 Amps. Consider the parallel connected transformer secondaries of Figure 12.

FIGURE 12. PARALLEL CONNECTED SECONDARY WINDINGS



In Figure 12. the two primary windings are connected the same across 240 volts but the two secondary windings are now connected in a parallel combination with respect to their dot orientation.

As with Figure 11, the two secondary windings are rated at 12V, 2.5A each. The secondary terminal voltage will be the same at 12 Volts but for parallel connected windings the currents add. So for parallel

connected secondary windings, the output of Figure 12. is rated at 12 Volts, 5.0 Amps.

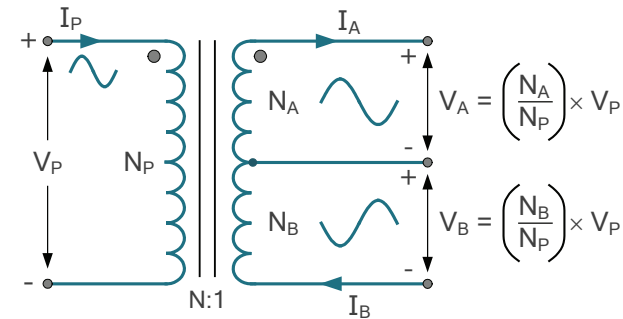
Transformer secondary windings must be correctly connected together to produce the required voltage or current output. Dot orientation is used on the windings to indicate the terminals that have the same phase relationship.

If not, connecting two secondary windings together in opposite dot-orientation will cause the magnetic flux created by the windings to cancel each other out resulting in damage to the transformer due to overheating.

Another type of dual voltage transformer which has only one secondary winding that is “tapped” at its electrical centre point is called a *centre-tapped* transformer.

Figure 13. shows the connection of a typical centre-tapped transformer. The tapping point is in the exact centre of the secondary winding providing a common connection for two equal but opposite secondary voltages.

FIGURE 13. CENTRE TAPPED TRANSFORMER



Here, the full secondary voltage would be the sum of V_A and V_B as for a single secondary coil.

However, if the centre-tapping point is used, the voltage output V_A would be positive in nature with respect to the centre tap, while the voltage at the other secondary, V_B will be negative and opposite polarity. That

is, they are both 180° electrical degrees out-of-phase with each other.

One disadvantage of using an ungrounded centre tapped transformer is that it can produce unbalanced voltages in either of the two secondary windings due to unsymmetrical currents flowing in the common centre tapped connection if the loads connected to secondary A and secondary B are unbalanced.

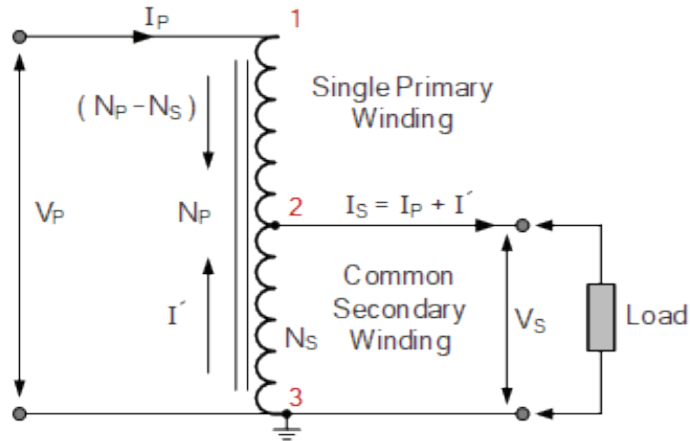
We could also produce a centre-tap transformer using the dual voltage transformer from Figure 11. By connecting the two secondary windings in series, we can use the centre link as a common tapping point. Thus, if the output from each secondary side is V , then the total output voltage for the secondary winding will be equal to $2V$.

10. THE AUTOTRANSFORMER

Unlike the previous voltage transformer which has two electrically isolated windings, the primary and the secondary, an autotransformer has only one single voltage winding which is common to both sides.

This single primary winding can be “tapped” at various points along its length to provide a percentage value of the primary voltage to the secondary load. Thus an autotransformer has the usual magnetic core but only has one winding, which is common to both the primary and secondary circuits as shown in Figure 14.

FIGURE 14. AN AUTOTRANSFORMER



The primary and secondary windings of an autotransformer are linked closely together both electrically and magnetically onto the same magnetic core. When the primary current, I_P is flowing through the single winding in the direction of the arrow in Figure 14, the secondary current, I_S flows in the opposite direction.

Thus, the portion of the common winding that generates the secondary voltage, V_S the current, I' flowing out of the winding is the difference between I_P and I_S . Thus depending on the load conditions, I' can be greater than I_P .

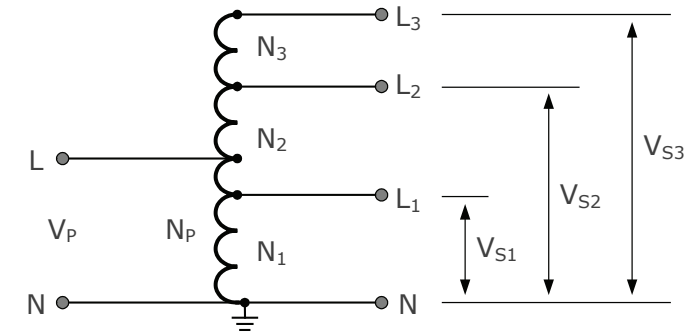
The advantage of an autotransformer design is that it can be made a lot cheaper for the same VA rating as less copper windings are used.

Autotransformers offer no electrical isolation between input and output

However, the main disadvantage of an autotransformer is that it does not have the primary/secondary winding isolation of a conventional double wound transformer. Thus any electrical fault in the single winding can result in the full primary voltage at the secondary terminals.

Autotransformers can also be constructed with more than one single or variable tapping point. They can also provide different voltage points along its main primary winding or increase a voltage level with respect to the applied voltage, V_P as shown in Figure 15.

FIGURE 15. AUTOTRANSFORMER WITH MULTIPLE TAPPING POINTS



The section designated as the primary part of the winding is connected to the AC power source with the secondary taps being part of this primary winding. An autotransformer with multi-tapping's can be used to step the applied voltage up or down by selecting the appropriate tapping connection. For example, secondary voltage V_{S2} and V_{S3} of Figure 15. would be greater than the applied voltage, L .

11. 3-PHASE TRANSFORMERS

Thus far we have looked at the construction and operation of single-phase, two winding voltage transformers. But transformers can also be constructed for connection to poly-phase supplies which use three or more alternating voltages and currents.

Three-phase, also written as 3-phase or 3ϕ supplies, are used for electrical power generation, transmission, and distribution, as well as for industrial uses. Three-phase supplies have many electrical advantages over single-phase power in that it can deliver more constant power to a load than the pulsating power of a single-phase supply.

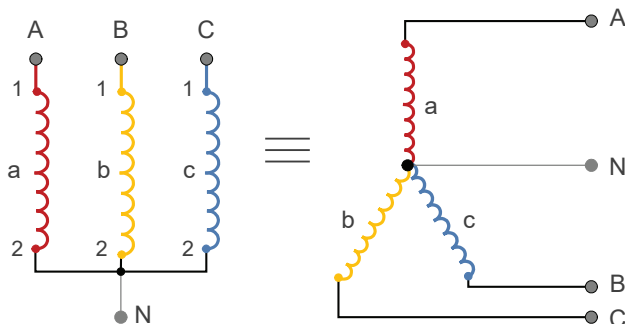
A three phase transformer is basically consists of three single-phase transformers wound onto a single magnetic core and therefore has three sets of primary and secondary windings, six in total with one for each phase.

Depending upon how these three sets of windings are interconnected, determines whether the transformer has a 4-terminal Star or Wye (Y) configuration, or a 3-terminal Delta or Mesh (Δ) configuration.

11.1 The Star or Wye (Y) Configuration

As its name implies, the “star” or “wye” configuration has one end of the three windings of the transformer connected to a common star point. This common star point is commonly referred to as the neutral point (N) since it is usually connected to ground. Generally, the three transformer windings use the colours Red, Yellow and Blue with capital (upper case) letters A, B and C, to represent the three individual windings as shown in Figure 16.

FIGURE 16. 3-PHASE STAR (WYE) CONNECTION



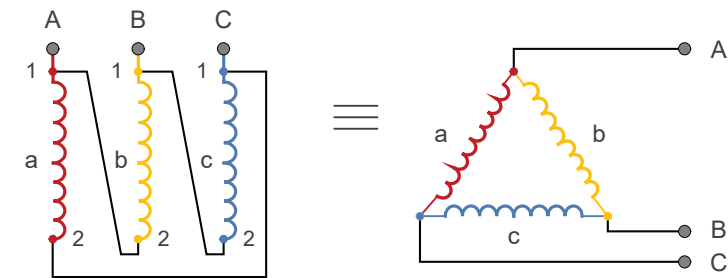
A three-phase star-connected transformer comprises of three line connections and a fourth neutral connection. The voltage between any line of the three-phase transformer is called the “line voltage”, V_L , while the voltage between any line and the neutral point of a star connected transformer is called the “phase voltage”, V_P .

Thus the three line-to-line voltages V_{AB} , V_{AC} and V_{BC} are or 1.732 times the nominal voltage of a single-phase transformer. That is: line voltage = $\sqrt{3}$ x phase voltage, or $V_L = \sqrt{3} \times V_P$. Also, since the current in one phase flows in through the line, the phase current, I_P of a star connected transformer is equal to the line current, I_L . That is: phase current equals the line current, or $I_P = I_L$.

11.2 The Delta (D) or Mesh (Δ) Configuration

For a “delta” or “mesh” configuration, one end of the three windings of the transformer is connected to the start of the next winding. Unlike the previous star configuration, the delta or mesh configuration does not have a fourth neutral connection, and can therefore be used with balanced loads as shown in Figure 17.

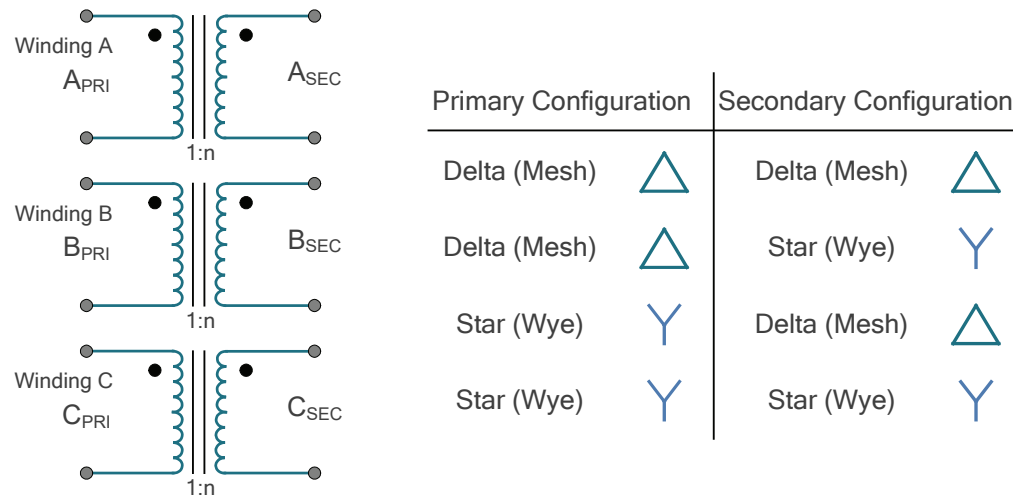
FIGURE 17. 3-PHASE DELTA (MESH) CONNECTION



As there is no fourth neutral connection, therefore a delta (mesh) connection has no phase voltage only line voltage, so the phase voltage equals the line voltage, or $V_P = V_L$. However, for a balanced delta connected load, the phase current in one phase of the load is defined as being: line current = $\sqrt{3}$ x phase current, or $I_L = \sqrt{3} \times I_P$.

The advantages of building a single 3-phase transformer onto one single laminated core is that for the same kVA rating it will be smaller, cheaper and lighter than three individual single phase transformers connected together because the copper and iron core are used more effectively.

The methods of connecting the primary and secondary windings are the same, whether using just one single 3-phase transformer or three separate single-phase transformers. Therefore, the primary and secondary windings of a three-phase transformer can be electrically connected in four possible ways as shown in Figure 18.

FIGURE 18. 3-PHASE TRANSFORMER CONNECTION


The secondary voltages of a three-phase delta-delta (Dd) or star-star (Yy) connected transformer depends only on the turns ratio, as potentially the primary and secondary voltages and currents would be of the same value for a 1:1 transformer.

However, if the 3-phase transformer is connected in star-delta, (Yd) configuration, each star-connected primary winding will receive the phase voltage, V_p of the supply, which is equal to: $1/\sqrt{3} \times V_L$. Thus each corresponding secondary winding would have this same voltage induced in it.

Since the secondary is delta-connected, the voltage ($1/\sqrt{3} \times V_L$) will become the secondary line voltage. Then with a 1:1 turns ratio, a star-delta connected transformer will provide a $\sqrt{3}:1$ step-down line-voltage ratio. Likewise, for a delta-star (Dy) connected transformer, with a 1:1 turns ratio, the transformer would provide a $1:\sqrt{3}$ step-up line-voltage ratio.

The relationship between the secondary line and phase voltages and currents in the four basic configurations of a three-phase transformer shown in Figure 18 with respect to the primary line voltage, V_L and its primary line current I_L as given in the following table.

TABLE 1. THREE-PHASE TRANSFORMER LINE VOLTAGES AND CURRENTS

	Line Voltage (V_L)	Line Current (I_L)	Phase Voltage (V_p)	Phase Current (I_p)
Star	$V_L = \sqrt{3} V_p$	$I_L = I_p$	$V_p = V_L / \sqrt{3}$	$I_p = I_L$
Delta	$V_L = V_p$	$I_L = \sqrt{3} I_p$	$V_p = V_L$	$I_p = I_L / \sqrt{3}$

Where: V_L is the line-to-line voltage with V_p being the phase-to-neutral voltage. Note that the power per phase is given as: $P = V_p I_p \cos\theta$, with the total power consumed by a three-phase load being equal to the sum of the amounts of power in each phase.

[End of this Alternating Current Transformers eBook](#)

Last revision: July 2022

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Main Headquarters

245 Main Street
Cambridge, MA 02142

www.aspencore.com

Central Europe/EMEA

Frankfurter Strasse 211
63263 Neu-Isenburg, Germany

info-europe@aspencore.com